COM668 Computing Project

Assessment Task 2 – Challenge Definition Report

Project title: **Cow Tag Information Management System**

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Contents

1. Problem definition 1
1.1. Aim 1
1.2. Problem description 1
1.3. Scope 1
1.4. Intended outcome 2
1.5. Objectives 2
2. Contextual research 2
2.1. Context investigation 2
2.2. Supporting theory research 3
2.3. Similar products review 5
3. Software definition 5
3.1. Requirements analysis 5
3.2. Risk analysis 7
3.3. Design 8
4. Planning 9
4.1. Methodology 11
4.2. Initial plans 12
References 15
Appendices 18

1. Problem definition

1.1. Aim

The goal of this project is to create cloud-based Cow Tag Information Management System (CTIMS) that lets farmers store, receive, and manage livestock data. This will use using unique electronic identification (EID) tag numbers which are used to register cows. The system will aim to provide a modern cattle management site which provides more accuracy, allowing farms to comply with regulations, run more efficiently and prevent data loss.

1.2. Problem description

The management of livestock is a massive problem for small/medium sized farms, which still use paper based and manual record keeping. These methods cause data loss, inconsistencies, duplication and limited access to key livestock operations, which makes farm operations less reliable (FAO,2023). Old methods also make it hard to track animal health reports, breeding cycles/relations and keep up to date with compliance records.

Digital traceability is increasingly important because of rule like the UK's Animal Health and Welfare Pathway (DEFRA, 2023), the EU's Animal Health Law (Regulation 2016/429), and the Food Standards Agency (2023) expectations for traceability of animals in the food supply chain. All these rules say farms must keep accurate, up to date and accessible records for their livestock. If they do not comply, they can be fined and have movement restrictions enforced.

CTIMS will need to follow UK GDPR rules for security, as it keeps track of user account information and identifiers for each farm. This includes secure authentication, encrypted data storage, and access controls (United Kingdom, 2018; NCSC, 2023). Farmers will be more likely to be hacked and have data stolen if they do not have a secure, centralized system.

The absence of an accessible, usable, and secure digital platform that supports accurate livestock record management while accommodating the practical constraints of small/medium-scale farming operations.

1.3. Scope

This project focuses on delivering a functional project for CTIMS with the following core capabilities:

- Create, view, update, and delete cow profiles linked to unique tag numbers.
- Record livestock data including health events, vaccinations, breeding records, and production information (FAO, 2023).
- Support rapid data retrieval using search and filtering tools.
- Provide dashboard-based summaries and analytics to support rapid interpretation of herd performance (W3C, 2021).
- Enforce secure user authentication aligned with cyber security best practice (NCSC, 2023).
- Store information securely within a scalable relational database system (Microsoft, 2024; W3Schools, 2024).

Out of scope (to constrain project size): RFID hardware integration, mobile apps, automated disease prediction, multi-farm administration, and integration with commercial herd management platforms.

1.4. Intended outcome

The intended outcomes include:

- A functional, web-based project that demonstrates secure authentication, CRUD operations, and a dashboard.
- A responsive and user-centred interface that supports common farming tasks with minimal cognitive load (Interaction Design Foundation, 2023).
- Reliable and scalable back-end infrastructure supporting data integrity and efficient query performance (Microsoft, 2024).
- A structured set of documents detailing requirements, data structures, design artefacts, and testing evidence to support further system development.
- Improvements in traceability, herd monitoring, and operational efficiency consistent with digital agriculture best practice (FAO, 2023).

1.5. Objectives

The SMART-aligned objectives for CTIMS are:

1. Review existing livestock management systems and digital trends to establish best practices (Herdwatch, CowManager, SUM-IT).
2. Analyse legal, ethical, and social considerations influencing system requirements (Food Standards Agency, 2023; DEFRA, 2023).
3. Identify stakeholders and define functional and non-functional requirements aligned with regulatory and data standards (BrowserStack, 2025).
4. Apply Agile methodology to structure iterative development and early testing (Atlassian, 2024).
5. Design and implement a secure relational database with enforced referential integrity (W3Schools, 2024; Microsoft, 2024).
6. Prototype a user-friendly, accessible interface aligned with recognised usability principles (Interaction Design Foundation, 2023; W3C, 2021).
7. Conduct testing to validate system functionality, performance, and usability (Cloudflare, 2024).
8. Evaluate the project critically and propose future enhancements based on findings.

2. Contextual research

2.1. Context investigation

Livestock farming is a key component in agriculture, food production, rural economies and international trade. The effective management of livestock relies on accurate, timely and accessible livestock records, ensuring animal welfare, productivity and maintain traceability across supply chains (FAO, 2023). As small/medium scale farms begin to follow these more complex and strict rulings, the paper-based and manual recording become inadequate to fulfil the requirements. They often lead to fragmented data, missing records, and difficulty meeting inspection requirements.

Modern policies reinforce the need for digital solutions. Within the UK, *Animal Health and Welfare Pathway*, promotes the monitoring of diseases and requires transparent, auditable livestock records (DEFRA, 2023). Across *Europe, Animal Health Law (Regulation 2016/429)* mandates all livestock must have accurate identification, allow for tracking of livestock movement and reliable historical data for disease control (European Union, 2016).The Food Standards Agency emphasises robust traceability for food safety, requiring farms to provide accurate animal identification data during inspections (Food Standards Agency, 2023).

These regulations cause burdens and extra work for farmers who lack the suitable tools and knowledge to meet all the requirements. An easy, lightweight system can directly address this gap in tools and knowledge by providing a structured storage for livestock records. This allows for consistency for farmers with the legal and compliance frameworks of these regulations.

2.1.1 Industry Drivers for Digital Livestock Management

Some of the drivers for digital livestock is the growing focus on animal welfare. Welfare requirements require record keeping of previous health problems, vaccinations and recent veterinary treatments. Poor or lack of this data limits a farm's ability to sell and move cattle, as there is no evidence of the compliance to the welfare requirements (DEFRA, 2023).

There is also an increasing emphasis on traceability of livestock. This is due to supply chains expecting full visibility of all livestock's life cycles, such as movement, health and birth records. A digital system will help reduce administrative errors which can lead to threats of biosecurity (Food Standards Agency, 2023) and allow farmers to sell their livestock with ease.

Another driver is the demand for operational efficiency, as digital platforms allow for the streamlining tasks such as registration of treatments and reviewing breeding connections. The allows for more informed decision making and reducing time managing paper work.

These drives show why farmers increasingly require a digital system that is accessible, secure and especially tailored to their own scale of operation.

2.1.2 Legal and Regulatory Considerations

This project must also comply with multiple legal and regulatory obligations, such as the *Animal Health and Welfare Pathway* (DEFRA, 2023), which encourages farms to maintain an up to date and accessible records to support in welfare inspections and health planning for livestock.

The *EU Regulation 2016/429* (European Union, 2016) requires enforcement of disease reporting, accurate tagging and transparent movement tracking of livestock when a farm wants to sell and move livestock.

The *UK GDPR and Data Protection Act 2018* (United Kingdom, 2018), must be followed as although livestock itself is not personal data, the project will store login credentials, business identifiers and unique user data. This means secure authentication, encryption, and access controls are required to protect user information on the platform.

These legal considerations directly influence the system's design and requirements. For example, strong authentication (NCSC, 2023) and traceable record updates support both GDPR obligations and farm-audit processes.

2.1.3 Ethical and Social Considerations

With building this project, ethical and social considerations must be considered, such as data ownership and trust, that farmers must feel confident that their data is stored both securely and not shared without consent. Having a clear ownership, being transparent and auditability is key for this project's creation (NCSC, 2023).

Many fall small farms operate in rural regions of the county, which may have unstable internet connection or farms may having limited use to technology. This means the project must prioritise a simple and efficient page loading to support the users (FAO, 2023).

The project must support proactive welfare monitoring by enabling fast identification of health issues through regular treatment and breeding history. Improved data visibility allows for earlier interventions of any health issues , benefiting the livestock welfare overall (DEFRA, 2023).

Overall, the analysis of the ethical and social considerations highlights the legal, ethical, operational, and technological environment that shapes the project. It also demonstrates a clear need for simple, accessible, and compliant digital solutions for livestock management.

2.2. Supporting theory research

2.2.1. Information Management and Database Theory

The structured storage and retrieval of livestock data is based on relational database theory. Normalising tables gets rid of extra data, makes sure that related records are consistent, and cuts down on mistakes when entering data, which is very important for traceability and compliance (W3Schools, 2024).

Entity–relationship modelling gives a conceptual picture of cows, health events, breeding records, and user accounts. This helps keep data organised logically and ensures referential integrity (Microsoft, 2024). These rules make sure that the historical data for each cow is linked correctly through foreign keys, which makes sure that queries and audits work reliably.

2.2.2. User-Centred Design and Usability Theory

User-centred design (UCD) helps make interfaces that are easy to use, predictable, and responsive to what users need. CTIMS uses UCD by making sure that page layouts are always the same, that navigation routes are clear, and that users get instant feedback when they do things like add a new record (Interaction Design Foundation, 2023).

Accessibility guidelines, like the WCAG 2.1 recommendations, help people choose colours, font sizes, and layout options that work for a wide range of users on different devices (W3C, 2021). This is especially important in farming areas where different devices and internet speeds are available.

2.2.3. Decision-Support and Data-Visualisation Theory

Dashboard design principles stress the importance of clarity, low cognitive load, and short visual cues to help people make decisions quickly. CTIMS uses summary cards and charts to show herd statistics, recent treatments, and breeding results. These kinds of visualisation techniques help farmers understand what's going on and make smart management choices (Development Gateway, 2025; W3C, 2021).

2.3. Similar products review

Herdwatch offers comprehensive herd management, breeding tools, and compliance reporting. However, its subscription cost and reliance on stable internet access make it less suitable for small-scale farmers seeking lightweight solutions (Herdwatch, n.d.).

CowManager provides highly detailed behavioural and fertility monitoring using sensor technology. While powerful, it requires specialised hardware and significant investment, limiting accessibility for smaller farms (CowManager, n.d.).

SUM-IT Farm Management delivers robust livestock and business management functionality but is often regarded as overly complex for farmers requiring only simple tag-based record management (SUM-IT, n.d.).

Across these systems, there is many gaps in the services they provide, such as many offers extensive features, which could overwhelm the user, who wants a simpler workflow. Many of these sites. Existing systems target larger farms and include complex features. CTIMS simplifies workflows, prioritising essential CRUD, traceability, and compliance to support users with limited digital literacy.

3. Software definition

3.1. Requirements analysis

3.1.1 Stakeholder Requirements

Farmers require quick access to livestock information via unique tag numbers, simple data-entry forms, and low-cognitive-load interfaces to support operational decisions.

Veterinarians require accurate health histories, vaccination details, and treatment notes for diagnosis and welfare management.

Farm managers require secure authentication, accountability through system logs, and reliable storage to meet regulatory obligations and audit expectations.

These needs directly inform the requirement set in Appendix A and Appendix B.

3.1.2 Functional Requirements - Referenced from Appendix A

The functional requirements listed in Appendix A (FR1–FR18) define the operational behaviours CTIMS must support.

- User authentication and session control (FR1–FR3) ensuring only authorised access to livestock records, supporting GDPR-aligned data protection expectations (United Kingdom, 2018)

- Comprehensive CRUD capability for cows, health records and breeding records (FR4–FR12), supporting traceability requirements within agricultural legislation (DEFRA, 2023).

- Search and filtering features (FR8–FR9) designed to reduce retrieval times and support decision-making during routine farming work.

- Dashboard-level summaries (FR13) providing herd-level insights consistent with usability guidance encouraging efficient interpretation (W3C, 2021).

- System logging, validation and error handling (FR14–FR17), ensuring accountability, data quality and operational reliability.

- Responsive interface behaviour (FR18) supporting access on desktop, tablet and farm office environments.

3.1.3 Non-Functional Requirements - Referenced from Appendix B

The non-functional requirements listed in Appendix B (NFR1–NFR13) set the system's quality criteria across usability, security, performance and maintainability

- Usability and accessibility (NFR1–NFR2) ensure the UI meets WCAG 2.1 recommendations (W3C, 2021) and can be used effectively by users with limited digital literacy (FAO, 2023).

- Performance and scalability (NFR3–NFR4) define acceptable response times and the capacity to store an expanding dataset as herd size grows.

- Reliability, integrity and compliance (NFR5–NFR8) ensure the system adheres to Azure SQL availability, follows GDPR principles, and maintains accurate, consistent records.

- Maintainability and portability (NFR9–NFR10) ensure future extension and flexible deployment environments.

- Backup, recovery and auditability (NFR11–NFR12) provide traceability and protection against data loss.

- Error tolerance (NFR13) ensures that user mistakes are handled gracefully.

3.1.4 Testing Plan

Testing will be requirement-driven and structured. Unit tests will validate individual modules (authentication, CRUD functions), integration testing will verify communication between the frontend, backend, and Azure SQL database, and User Acceptance Testing (UAT) will confirm usability and stakeholder satisfaction. Security testing will include input validation and authentication checks aligned with secure development practices (Cloudflare, 2024).

Each test case is explicitly mapped to its corresponding functional or non-functional requirement to ensure full traceability and validation coverage, supporting best practice quality assurance in system development (BrowserStack, 2025).

3.2. Risk analysis

A formal risk assessment was conducted using likelihood–impact scoring (1–5 scale). Each risk is aligned to the requirements in Appendix A and Appendix B, following secure development guidance (NCSC, 2023; Cloudflare, 2024).

3.2.1 Technical Risks

Risk ID	Description	Likelihood	Impact	Score	Related Requirements	Mitigation Strategy
R1	Data loss/corruption	3	5	15	NFR7, NFR11, FR15	Azure SQL backups; referential integrity; transactional design
R2	Security breach	2	5	10	FR1–FR3, NFR6	Password hashing; input validation; encrypted communications
R3	Poor performance under data growth	3	4	12	NFR3, NFR4, FR13	Query optimisation; indexing tag_number; appropriate Azure tier

3.2.2 Operational Risks

Risk ID	Description	Likelihood	Impact	Score	Related Requirements	Mitigation
R4	Incorrect data entry	4	3	12	FR16, FR17, NFR13	Validation rules; dropdown menus; guided forms
R5	Limited rural connectivity	3	3	9	NFR1, NFR3, FR18	Lightweight UI; optimised queries; compressed static assets

3.2.3 Project Risks

Risk ID	Description	Score	Mitigation
R6	Over-ambitious scope	12	MoSCoW prioritisation based on FRs/NFRs
R7	Insufficient testing time	12	Embedded testing per sprint; test prioritisation

Additional risks include authentication failure and Azure cost escalation. Mitigations include session-expiry controls, secure credential handling, and Azure cost-alert monitoring to avoid budget overruns.

3.3. Design

3.3.1. System Design

CTIMS uses a three-tier architecture deployed on Microsoft Azure, selected due to its reliability, automatic scaling and built-in security controls (Microsoft, 2024). The architecture supports the functional needs (FR1–FR18) and non-functional criteria (NFR1–NFR13). Flask was selected over Django and FastAPI due to its lighter footprint and suitability for focused CRUD workflows. Azure SQL was chosen instead of NoSQL alternatives to ensure relational integrity and auditability. A three-tier architecture improves maintainability, security boundaries, and supports NFR5 and NFR6.

Presentation Layer

- HTML, CSS, JavaScript
- Implements UI, dashboards, search and CRUD interfaces which supports FR5–FR13 and NFR1–NFR3
- Hosted on Azure App Service for HTTPS, monitoring and rapid deployment

Application Layer

- Python Flask backend
- Processes authentication, validation (FR16), error handling (FR17) and logging (FR14)
- Applies security measures recommended by NCSC (2023) and Cloudflare (2024)

Data Layer (Azure SQL Database)

- Enforces unique tag numbers (FR15) and referential integrity (FR7)
- Provides automatic failover, geo-replication (NFR11), and high availability (NFR5)

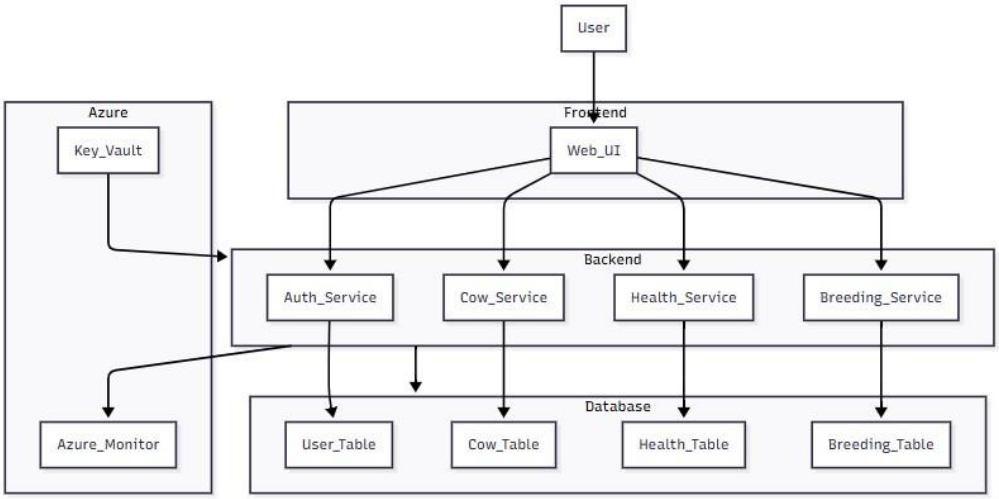

Figure 1: CTIMS Three-Tier Cloud Architecture showing the flow between UI, Flask backend and Azure SQL Database (MermaidJS, 2024).

3.3.2 UML Use Case Design

The UML Use Case Diagram models essential interactions such as registration, login, CRUD operations, search, filtering and dashboard access (FR1–FR18). <<include>> relationships prevent redundancy and ensure consistent reuse of behaviours such as “Search by Tag Number”.

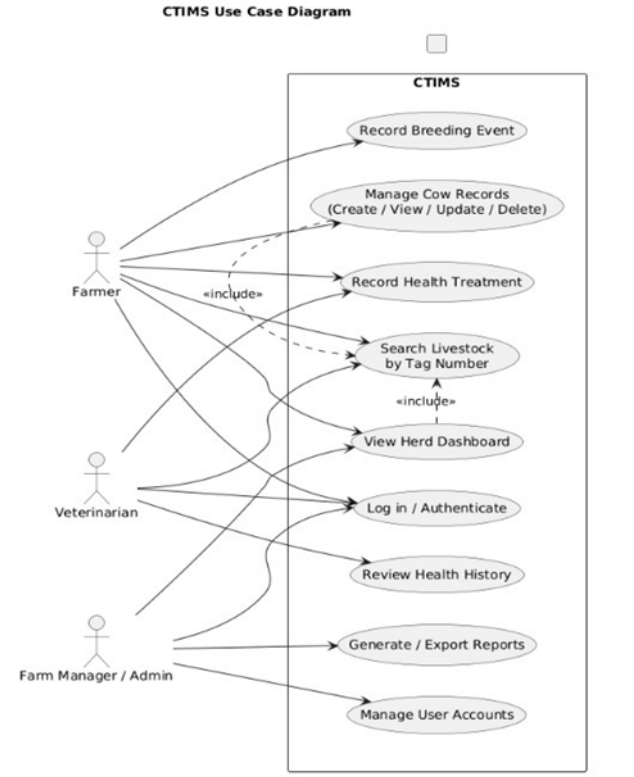

Figure 2 : CTIMS UML Use Case Diagram showing user interactions and shared behaviours (PlantUML, 2025).

3.3.3 Data Design

CTIMS uses a relational database design aligned with W3Schools (2024) and Microsoft (2024) recommendations. It supports:

- Unique identifiers
- Normalised tables
- Referential integrity
- Efficient query performance (indexing tag_number)

Core Tables

User Entity

- user_id: Integer, primary key. Uniquely identifies each registered user.
- username: String, unique, required. Used for login and account identification.

- hashed_password: String, required. Contains securely hashed user credentials in line with NCSC (2023).

- role: String. Indicates user permissions (e.g., farmer, vet, admin) and supports access control.

Cow Entity

- cow_id: Integer, primary key. Internal identifier for each cow.
- tag_number: String, unique, required. Official livestock identifier used for traceability.
- breed: String. Records the cow's breed for management purposes.
- sex: String. Indicates whether the cow is male or female.
- location: String. Specifies the cow's housing or grazing location.
- date_of_birth: Date. Records birth date for age and lifecycle calculations.
- status: String. Indicates animal condition (e.g., active, sold, deceased).
- user_id: Integer, foreign key. Links the cow to its owner/manager in the User entity.

Health Record Enity

- health_id: Integer, primary key. Uniquely identifies each health event.
- cow_id: Integer, foreign key. Associates the health record with the correct cow.
- treatment_date: Date. When the vaccination or treatment occurred.
- vaccination_type: String. Name or type of vaccine or health intervention.
- notes: Text. Optional additional information supplied by the user or vet.

Breeding_Record Entity

- breeding_id: Integer, primary key. Uniquely identifies each recorded breeding event.
- cow_id: Integer, foreign key. Links the breeding entry to a specific cow.
- service_date: Date. The date the breeding service occurred.
- outcome: String. Describes the result (e.g., successful, failed, pending).

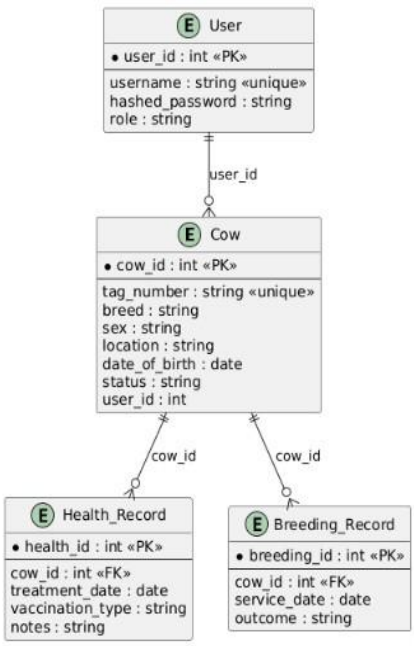

Figure 3: Entity–Relationship Diagram showing CTIMS entities and relationships (PlantUML, 2025).

3.3.4 User Experience Design (UX)

The UX design for CTIMS prioritises simplicity, consistency, and accessibility to support farmers with varied levels of digital literacy. Interfaces follow WCAG 2.1 guidance for contrast, readability, and scalable text (W3C, 2021), while maintaining consistent layouts, navigation patterns, and icons to minimise cognitive load. Each interaction provides clear visual feedback, ensuring users understand when records are added, updated, or contain errors.

Core screens include the login, dashboard, livestock list, and cow profile pages are designed for clarity and efficiency. The login page provides secure authentication with informative error messages, while the dashboard summarises herd size, recent treatments, and breeding activity using optimised Azure SQL queries. Livestock records can be searched or filtered and detailed cow profiles clearly separate health and breeding information. Record-entry forms use dropdowns, validation, and date pickers to reduce errors. Performance is supported through compressed assets, selective page updates, and efficient database queries.

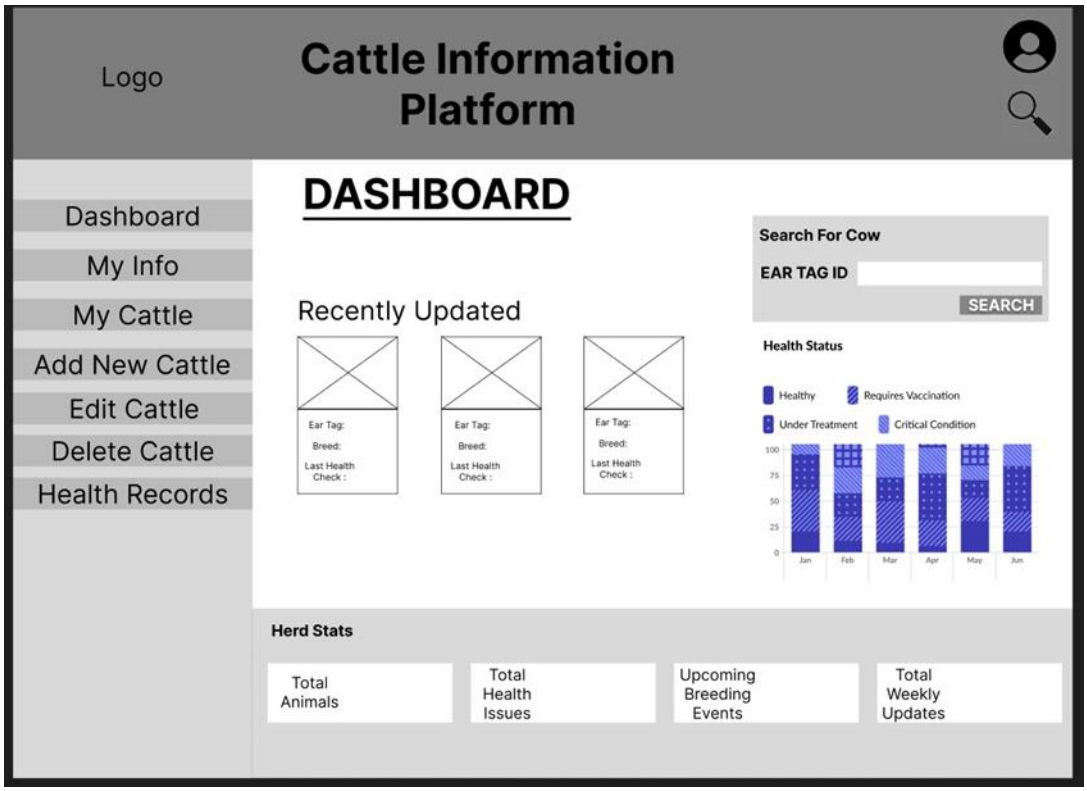

Figure 4: A sample dashboard created for the farmer . Figma (2025)

4. Planning

4.1. Methodology

The Cow Tag Information Management System (CTIMS) will be developed using Scrum principles. Agile was chosen for its flexibility, user-centeredness, and suitability for projects where requirements may change as system understanding deepens (Atlassian, 2024; GitHub, 2025). Agile allows continuous feedback, incremental delivery, and early issue identification, which are necessary to build a functional and user-friendly system in an academic timeframe.

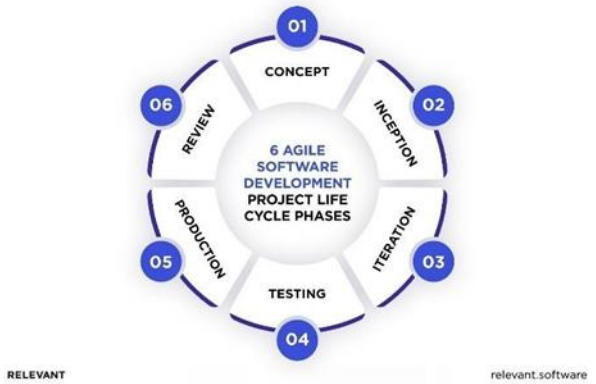

Figure 5: Agile Development Lifecycle
Source: Relevant Software (2023)

While fully linear methodologies such as Waterfall were considered, their rigid, sequential structure was not suitable for CTIMS because the requirements evolved throughout development. However, academic assessment criteria require formal documentation— such as requirement specifications, design diagrams and planning artefacts—which are typically associated with more structured approaches.

For this reason, a hybrid Agile approach was adopted: Agile was used for iteration, feedback, and incremental development, while structured documentation was included solely to meet academic standards, rather than as a Waterfall style process model.

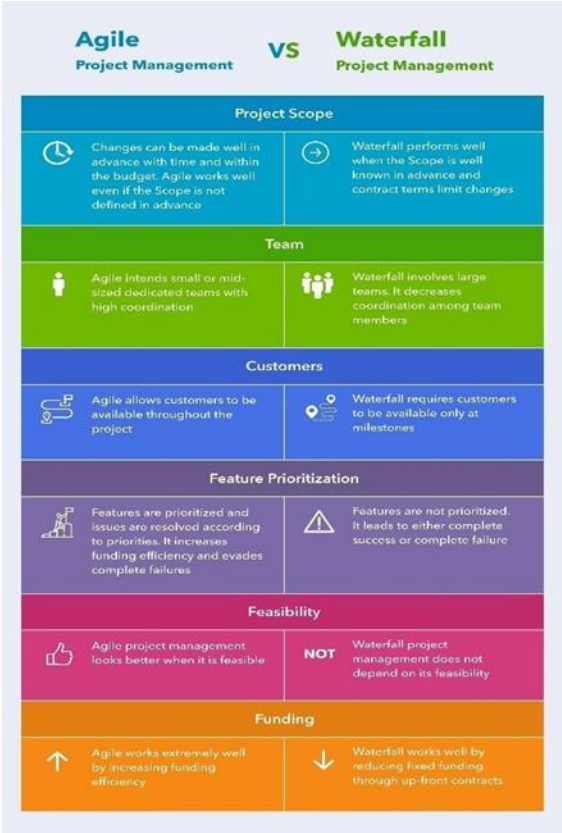

Figure 6. Comparison of Agile and Waterfall Methodologies
Source: DCM Learning (2024)

The methodology will incorporate the following core Agile practices:

Sprint-based development:

- The project will be divided into short development cycles (sprints), each focusing on a specific feature set (e.g., user login, CRUD functionality, dashboard display).

Incremental delivery:

- Features will be implemented in priority order, allowing early demonstration of system capabilities and progressive system maturity.

Continuous testing and feedback:

- Each sprint will include functional testing and performance checks, ensuring issues are detected early and resolved efficiently.

A hybrid approach will also be applied, combining Agile flexibility with structured planning documentation (e.g., requirement specification and design documents) to satisfy academic assessment standards. This ensures traceability, consistency, and clear progress reporting while preserving iterative adaptability (GitHub, 2025).

The use of Agile aligns effectively with CTIMS's modular architecture, cloud-based deployment, and evolving feature set. It enables focused work on core functionalities such as CRUD operations, data validation, and dashboard analytics before moving to subsequent enhancements.

4.2. Initial plans

The initial planning phase will establish a structured development roadmap to guide the project from conception through implementation and evaluation. This plan will ensure efficient time management and alignment with academic deadlines.

Project Breakdown Structure
The project will be divided into the following high-level phases:

1. Research and Analysis
   - o Review literature and similar livestock management systems
   - o Define stakeholder needs and system requirements
2. System Design
   - o Create system architecture diagrams and ERDs
   - o Design wireframes and user interface layouts
   - o Define Azure configuration plans
3. Implementation
   - o Develop authentication system
   - o Implement CRUD operations for livestock data
   - o Integrate Azure SQL Database
   - o Build dashboard functionality
4. Testing and Validation
   - o Unit testing of individual components
   - o Integration testing across system layers
   - o User Acceptance Testing (UAT)
5. Evaluation and Documentation
   - o Compare system performance to objectives
   - o Identify limitations and future improvements
   - o Produce final technical documentation

Timeline Strategy

To manage these phases effectively, a Gantt chart will be used to illustrate task scheduling, deadlines, and dependencies. Weekly milestones will help monitor progress and ensure that any delays or deviations from the planned timeline are identified early (Atlassian, 2024).

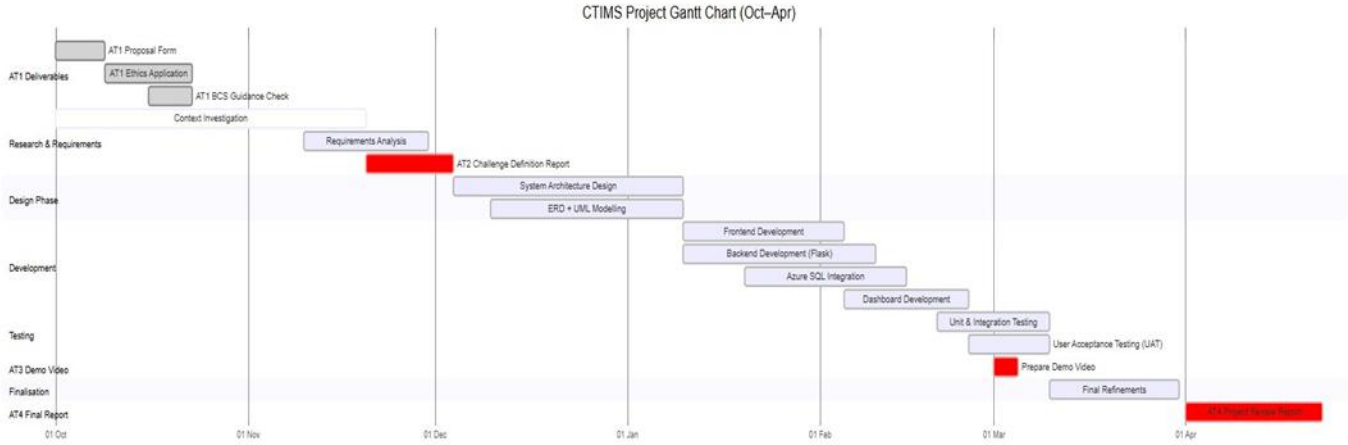

Figure 7: Gantt chart created using MermaidJS (2024).

Tool and Resource Planning

The following tools and technologies will support project execution:

- Development Environment: Visual Studio Code
- Backend Framework: Python Flask o Database: Azure SQL Database
- Hosting Platform: Azure App Service
- Version Control: GitHub
- Testing Tools: PyTest, Postman and manual testing scripts

These resources ensure compatibility with modern development standards and support the implementation of a secure, scalable web application (Microsoft, 2024; GitHub, 2025).

Risk-aware Planning

The plan also accounts for potential challenges such as time constraints or technical issues by incorporating buffer periods within the schedule. This allows flexibility for debugging, system refinement, and additional testing where necessary.

Summary

The selected Agile methodology and structured planning approach ensure CTIMS development remains organised, adaptable, and aligned with both academic and industry standards. By combining iterative development with detailed scheduling and documentation, the project is positioned to deliver a robust, user-centred livestock management system within the specified timeframe.

Critical Reflection and Professional Justification

This project shows that modern software engineering closely matches agricultural challenges. Cloud services like Azure SQL and Azure App Service improve reliability, security, and scalability over on-premise systems, supporting compliance, traceability, and data integrity (Microsoft, 2024). Integrating Agile methodology ensures adaptability and continuous refinement, meeting professional development standards (Atlassian, 2024). CTIMS addresses technical and operational limitations in livestock management systems to balance usability, performance, and future scalability.

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Appendices

Appendix A — Functional Requirements (FR)

ID	Functional Requirement	Description / Expected Behaviour
FR1	User Registration	The system must allow new users to create an account with secure credential storage.
FR2	User Login	Only authenticated users can access CTIMS features. Incorrect attempts must be handled securely.
FR3	User Logout	Users must be able to log out, ending the active session.
FR4	Create Cow Record	Users can create a new cow profile containing tag, breed, sex, location, DOB, and status.
FR5	View Cow Record	Users can view all details of an individual cow and linked records.
FR6	Update Cow Record	Users can edit existing cow data including health and breeding information.
FR7	Delete Cow Record	Users can delete a cow profile, ensuring referential integrity for linked records.
FR8	Search by Tag Number	Users can search for a cow using its unique tag or tag number.
FR9	Filter Cow Records	Optional filters for breed, age, health status, and location must be available.
FR10	Add Health Record	Users can record treatments, vaccinations, and health notes linked to a cow.
FR11	Update Health Record	Users can update previously added treatment or vaccination details.
FR12	Add Breeding Record	Users can record breeding activity, dates, and outcomes for each cow.
FR13	Dashboard Overview	System displays herd statistics such as total cows, recent treatments, and breeding alerts.
FR14	System Logs	System logs key user actions such as create, update, and delete events.
FR15	Database Storage	All data must be stored in Azure SQL with unique tag enforcement and referential integrity.
FR16	Data Validation	The system must validate input before submission (dates, formats, mandatory fields).

FR17	**Error Handling**	**The system must notify users of invalid actions, failed inputs, or missing fields.**
FR18	**Responsive Interface**	**The interface must adjust to different screen sizes (desktop, tablet, phone).**

Table B — Non-Functional Requirements (NFR)

ID	Non-Functional Requirement	Description / Quality Measure
NFR1	**Usability**	**Interface must be simple and intuitive, requiring minimal training for farmers.**
NFR2	**Accessibility**	**UI must meet WCAG 2.1 guidance for contrast, readability, and keyboard navigation.**
NFR3	**Performance**	**If connection is available, results should appear in 5 seconds**
NFR4	**Scalability**	**System must handle increasing numbers of records as herd size grows.**
NFR5	**Reliability**	**System must remain operational with minimal downtime; Azure SQL ensures high availability.**
NFR6	**Security**	**Passwords must be hashed; communications encrypted; OWASP risks mitigated.**
NFR7	**Data Integrity**	**All CRUD operations must maintain accurate, consistent data without duplication.**
NFR8	**Compliance**	**System must meet UK GDPR and livestock data regulations for secure information handling.**
NFR9	**Maintainability**	**Code should follow clean, modular design to support future enhancements.**
NFR10	**Portability**	**System must run in modern browsers without additional software requirements.**
NFR11	**Backup & Recovery**	**Automatic backups must protect against data loss; Azure SQL provides geo-replication.**
NFR12	**Auditability**	**System logs must allow tracking of key user actions for traceability.**
NFR13	**Error Tolerance**	**Minor input mistakes should prompt corrective feedback rather than system failure.**